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Forensics Lab
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September 11, 2023



GRADUATE DIGITAL FORENSIC REPORT



INTRODUCTION

The purpose of this report is to provide corroborative evidence of means, motive and opportunity leading to the fraud perpetrated by Senator Palpatine. The facts presented within this report are those within my area of expertise and knowledge and do not extend to matters outside of the same.

Summary of the case and tasking

- Between the period beginning January 2, 2022, and ending August 31, 2023, Senator Palpatine served as CEO for The Galactic Empire, LLC (“Company”), which managed various profit sharing and retirement plans. During this time, he coordinated a systematic exploitation scheme transferring the funds of the plans the Company manages to his personal accounts.
- As a result, both the Company and a substantial portion of its investors suffered monetary loss in excess of XXXXXX, with the Company becoming personally liable for all lost investments and tax liabilities associated with excessive early withdrawals.
- This fraudulent scheme has rendered the Company insolvent.
- On September 10, 2023, Detective Sergeant Ben Kenobi provided forensic images of both Senator Palpatine’s home desktop and work laptop. I was retained to reconstruct, to the extent presently possible, any documentary evidence reflecting the activities of Senator Palpatine which would corroborate professional negligence, misconduct, or fraud.

Statement of compliance

- I understand my duty as an expert witness to provide independent assistance by way of objective unbiased opinion in relation to matters within my expertise. I will inform all parties in the event that my opinion changes in any way material to this investigation.

FORENSIC EXAMINATION & ANALYSIS

All examinations, tests, measurements, and experiments related to the digital image of the accused computer were performed by me. Having booted the forensic workstation to Windows 11, 64-bit system with 64 GB RAM, 10.0.22621, I verified the forensic database using FTK Imager 4.7.1.2 as 0ac8cdf02c7062143db4283c08c5a39d. After mounting it with Arsenal Image Mounter 3.10.262 (figure 1-2) Following this procedure all requested items are satisfied with the utilization of the following forensic tools:

Tools

The forensics tools employed in the performance of this investigation were as follows:

σ **FTK Imager** – Exterro Version 7.6

Forensic software acquires electronic evidence in a forensically sound manner by creating copies of the computer data without making any changes to the original evidence.

σ **Autopsy** – Version 4.21.0

Autopsy can be used to analyze Windows and UNIX file systems, including NTFS, FAT, FFS, EXT2FS, and EXT3FS. It can also recover deleted files and show various sectors of uploaded images.

σ **Windows Calculator**

software calculator that comes pre-installed with Windows

σ **Volatility** – Version v1.0.0

Reads the Prefetch files stored in the system and displays the information stored in them. By looking in these files, we can learn which files every application is using, and which files are loaded on Windows boot.

σ **Google Search Engine**

works by continuously mapping the web and other sources to connect users to the most relevant information.

σ **Microsoft Excel**

Used for data entry and management, charts, and graphs, and convert JSON to CSV.

σ **Arsenal Image Mounter (AIM)** - Version v2.9.217

Mounts forensic image or virtual machine disk in Windows.

To my knowledge, there is no software in use above that possesses any material issues whose severity would invalidate the findings hereof.

Forensics Acquisition & Exam Preparation

In line with our objective, we focused on obtaining evidence to support investigation of Senator Palpatine's activities as well as the professional negligence of the External Auditor, which led to the pecuniary loss to the Company.

Following this procedure all requested items are satisfied with the utilization of FTK Imager 4.7.1.2, HxD, Autopsy 4.21.0, Arsenal Image Mounter 3.10.262, Microsoft Edge Browser, Notepad, and DB Browser for SQLite 3.12.3.

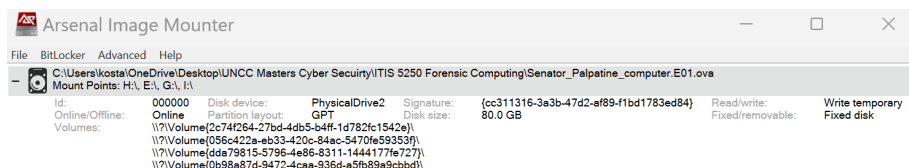


Figure 1: Mounting the forensic image using Arsenal.

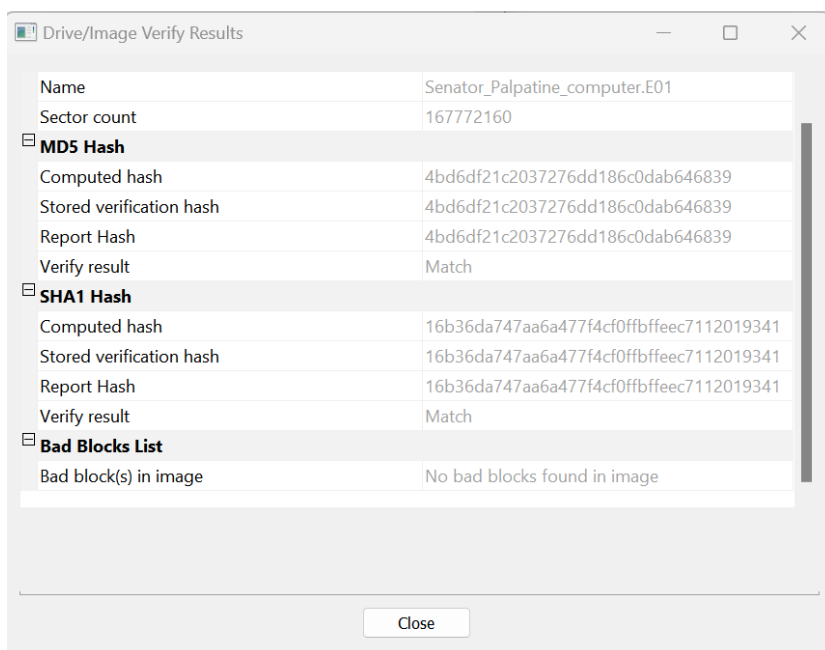


Figure 2: Verified MD5 Hash Values.

1. Account for how all the space on the laptop hard drive was used and determine the type of file system in use.

The Palpatine.E01 image has a byte per sector count of 512 and a sector count of 167,772,160. Using the Windows calculator, we found that the hard drive size is 85,899,345,920 or ~ 85,899,345.92MB. Using FTK Imager, we found that the image has four partitions and unpartitioned space. Partition 1 has a starting sector of 2,048 and a sector count of 614,400. Overall, it has a size of 300MB. Partition 2 has a starting sector of 616,448 and a sector count of 204,800. Overall, it has a size of 100MB. Partition 3 has a starting sector of 821,248 and a sector count of 262,144. Overall, it has a size of 128MB. Partition 4 has a starting sector of 1,083,392 and a sector count of 164,638,720. Overall, it has a size of 80390MB. Partition 5 has a starting sector of 165,722,112 and a sector count of 2,045,952. Overall, it has a size of 999MB. The unpartitioned space has no size or partition information.

Properties	
	
Partition Information	
Starting Sector	2,048
Sector Count	614,400
GUID Partition Table Information	
Partition Type GUID	{DE94BBA4-06D1-4D40-A16A-BFD5017}
Unique Partition GUID	{2C74F264-27BD-4DB5-B4FF-1D782FC1}

Figure 3: Partition 1 partition information and size using FTK Imager 4.7.1.2.

Properties	
	
Partition Information	
Starting Sector	616,448
Sector Count	204,800
GUID Partition Table Information	
Partition Type GUID	{C12A7328-F81F-11D2-BA4B-00A0C93E}
Unique Partition GUID	{0B98A87D-9472-4CAA-936D-A5FB89A}

Figure 4 Partition 2 partition information and size using FTK Imager 4.7.1.2.

Properties	
 2↓	
Partition Information	
Starting Sector	821,248
Sector Count	262,144
GUID Partition Table Information	
Partition Type GUID	{E3C9E316-0B5C-4DB8-817D-F92DF002}
Unique Partition GUID	{CD34EAB4-56C2-4860-9115-F4CDFABF}

Figure 5 Partition 3 partition information and size using FTK Imager 4.7.1.2.

Properties	
 2↓	
Partition Information	
Starting Sector	1,083,392
Sector Count	164,638,720
GUID Partition Table Information	
Partition Type GUID	{EBD0A0A2-B9E5-4433-87C0-68B6B726}
Unique Partition GUID	{056C422A-EB33-420C-84AC-5470FE59}

Figure 6: Partition 4 partition information and size using FTK Imager 4.7.1.2.

Partition Information	
Starting Sector	165,722,112
Sector Count	2,045,952
GUID Partition Table Information	
Partition Type GUID	{EBD0A0A2-B9E5-4433-87C0-68B6B726}
Unique Partition GUID	{DDA79815-5796-4E86-8311-1444177F}

Figure 7: Partition 5 partition information and size using FTK Imager 4.7.1.2.

Evidence Source Path	\\.\PHYSICALDRIVE2
Evidence Type	Physical Disk
Disk	
Drive Geometry	
Cylinders	10,443
Tracks per Cylinder	255
Sectors per Track	63
Bytes per Sector	512
Sector Count	167,772,160
Physical Drive Information	
Drive Model	Arsenal Virtual SCSI Disk Device
Drive Serial Number	{ca5f1d24-8fc4-11ee-af54-581cf83013}
Drive Interface Type	SCSI
Removable drive	False

Figure 8: image of disc geometry using FTK Imager

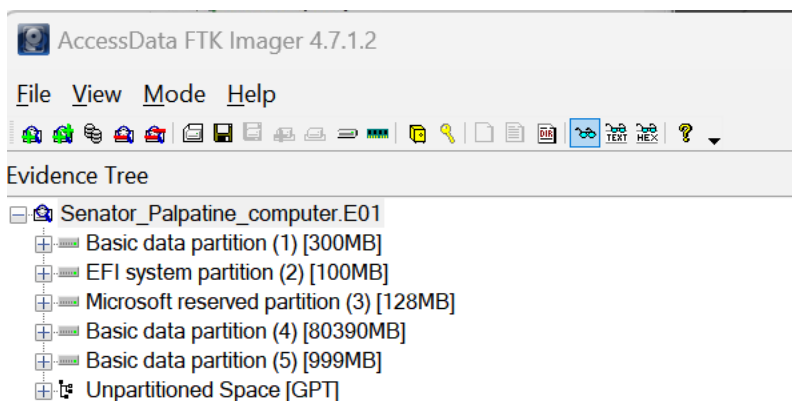


Figure 9: Unpartitioned Space using FTK Imager 4.7.1.2.

$$167772160 \times 512 = 85,899,345,920$$

Figure 10: Calculation of Image geometry using Windows calculator.

2. Identify the type of computer, version, and build number of the operating system.

By using Volatility 3 the Windows.info function, I was able to forensically analyze the memory dump of the RAM. Here we were able to glean that the operating system and build being used was Windows 10 32bit, as seen in figure 12.

```
Kernel Base 0xf80352600000
DTB 0x1ae000
Symbols file:///C:/Users/kosta/OneDrive/Desktop/UNCC%20Masters%20Cyber%20Secuirty/ITIS%205250%20Forensic%20C
duateProject/volatility3-2.4.1/volatility3/symbols/windows/ntkrnlmp.pdb/F9F3101286B6467CDE2D6C8304D7F43C-1.j
Is64Bit True
TePae False
layer_name 0 WindowsIntel32e
memory_layer 1 FileLayer
KdVersionBlock 0xf803532099a8
Major/Minor 15.22621
MachineType 34404
KeNumberProcessors 32
SystemTime 2023-11-29 03:23:20
NtSystemRoot C:\Windows
NtProductType NtProductWinNt
NtMajorVersion 10
NtMinorVersion 0
PE MajorOperatingSystemVersion 10
PE MinorOperatingSystemVersion 0
PE Machine 34404
PE TimeDateStamp Mon Dec 11 10:40:12 1989
PS C:\Users\kosta\OneDrive\Desktop\UNCC Masters Cyber Secuirty\ITIS 5250 Forensic Computing\GraduateProject\
2.4.1>
```

Figure 11: Ram memory dump showing operating system using Volatility 3.

3. Palpatine told investigators that “they had it all wrong”, the computer he was using was a Dell Alienware. Identify the type of computer the operating system was running on.

By using Volatility3, the “pslist” function which lists all running processes on the system, we were able to determine that the computer was in fact an Asus, not a Dell. (figure 13)

```
32668 11820 asus_framework 0x940b702c90c0 0 - 1 True 2023-11-29 00:39:25.000000 2023-11-
29 00:39:28.000000 Disabled
```

Figure 12: Computer type found in dump memory using Volatility3.

4. What type of file system was being utilized?

By utilizing the utility of FTK Imager we were able to discover two distinct files systems being used by Palpatine's computer, NYFS and FAT32. (figure 10)

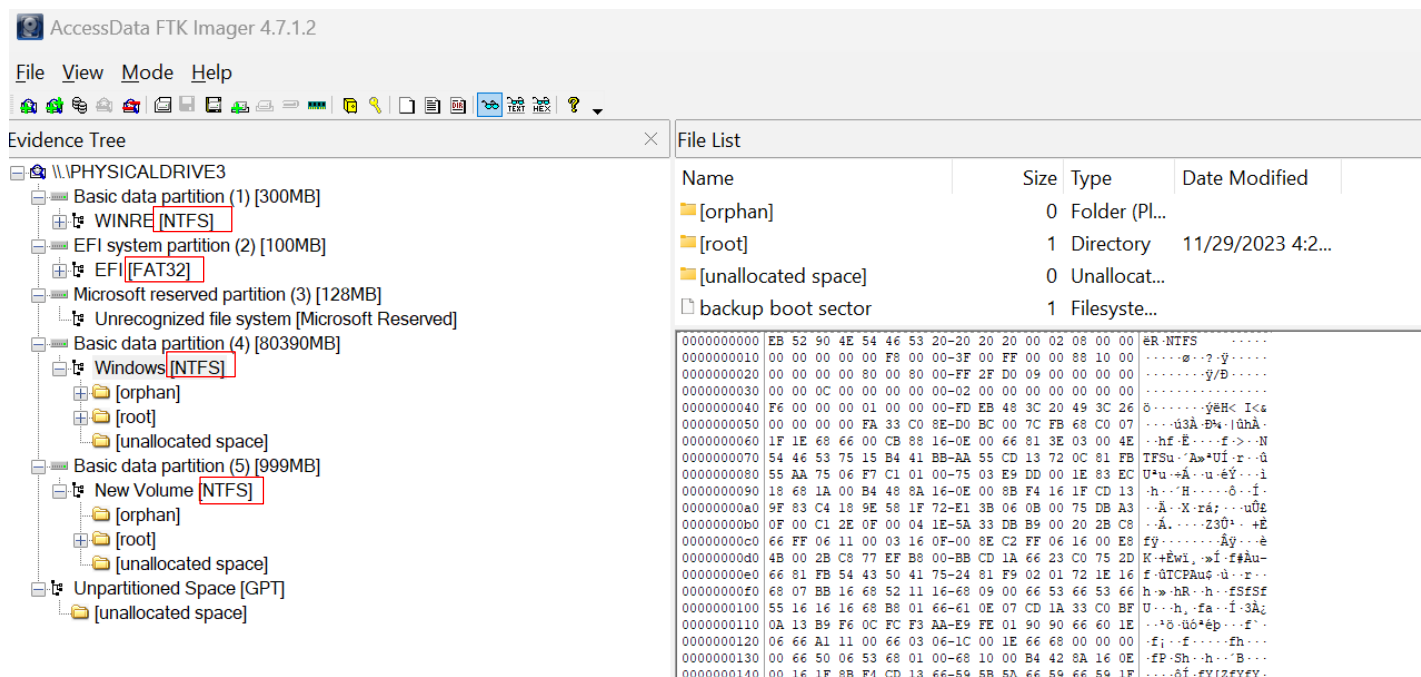


Figure 13: File system used in Palpatine's computer.

5. Were there any hidden files that may have revealed Palpatine's intent to commit fraud?

There were deleted files in Excel format that were carved from the hex values using HxD and Autopsy. One of which was entitled "ExpenseGe.xlsx". This file named a project, "Death Star" and the retirement accounts funding this project for over a year's projection. (figure 15-16)

Offset(h)	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F	Decoded text
00019730	00	00	00	00	00	00	00	00	C0	2F	FF	E3	7F	FC	FF	01	Ä/ÿä.ÿ.
00019740	35	8C	79	C3	64	65	B2	F4	00	00	00	00	49	45	4E	44	5GyÄde=ö....IEND
00019750	AE	42	60	82	50	4B	03	04	0A	00	00	00	00	00	00	00	0B`,PK.....
00019760	21	00	E4	25	8B	2F	A2	5D	00	00	A2	5D	00	00	13	00	!.ä%</c]...c]....
00019770	00	00	78	6C	2F	6D	65	64	69	61	2F	69	6D	61	67	65	..xl/media/image
00019780	32	2E	73	76	67	3C	73	76	67	20	76	69	65	77	42	6F	2.svg<svg viewBo
00019790	78	3D	22	30	20	30	20	32	35	32	2E	34	36	20	32	35	x="0 0 252.46 25
000197A0	36	22	20	78	6D	6C	6E	73	3D	22	68	74	74	70	3A	2F	6" xmlns="http:/
000197B0	2F	77	77	77	2E	77	33	2E	6F	72	67	2F	32	30	30	30	/www.w3.org/2000
000197C0	2F	73	76	67	22	20	78	6D	6C	6E	73	3A	78	6C	69	6E	/svg" xmlns:xlin
000197D0	6B	3D	22	68	74	74	70	3A	2F	2F	77	77	77	2E	77	33	k="http://www.w3
000197E0	2E	6F	72	67	2F	31	39	39	39	2F	78	6C	69	6E	6B	22	.org/1999/xlink"
000197F0	20	69	64	3D	22	61	22	20	6F	76	65	72	66	6C	6F	77	id="a" overflow
00019800	3D	22	68	69	64	64	65	6E	22	3E	3C	70	61	74	68	20	= "hidden"><path
00019810	64	3D	22	4D	32	33	35	2E	32	32	20	31	30	2E	39	43	d="M235.22 10.9C
00019820	32	33	34	2E	30	39	20	31	32	2E	30	37	20	32	33	34	234.09 12.07 234
00019830	2E	39	33	20	31	33	2E	38	35	20	32	33	35	2E	30	33	.93 13.85 235.03
00019840	20	31	35	2E	32	35	20	32	33	35	2E	31	37	20	31	37	15.25 235.17 17
00019850	2E	31	38	20	32	33	34	2E	39	35	20	31	39	2E	31	34	.18 234.95 19.14
00019860	20	32	33	34	2E	38	33	20	32	31	2E	30	36	20	32	33	234.83 21.06 23
00019870	34	2E	36	31	20	32	34	2E	35	38	20	32	33	34	2E	37	4.61 24.58 234.7
00019880	20	32	38	2E	30	35	20	32	33	34	2E	39	38	20	33	31	28.05 234.98 31
00019890	2E	35	37	20	32	33	35	2E	33	36	20	33	36	2E	34	20	.57 235.36 36.4
000198A0	32	33	35	2E	37	37	20	34	31	2E	31	39	20	32	33	37	235.77 41.19 235
000198B0	2E	38	30	34	36	3E	30	34	30	32	3E	38	31	30	31	30	8.46 04 235.81

Figure 14: Hidden file carved by HxD found in recycle.

Death Star PLANNED EXPENSES													DS-1 ORBITAL BATTLE STATION COST OF STEEL 1852,000,000,000,000,000 R&D INTERIORS, WEAPONRY 992,148,000,000,000,000,000 0.2% OF GDP GALACTIC EMPIRE GDP 54,600,000,000,000,000,000,000
	January	February	March	April		June	July	August	September	October	November	December	Total
RETIREMENT COSTS	\$ 50,000,000.00	\$ 66,000,000.00	\$ 40,085,000.00	\$ 50,087,500.00	*****	\$ 70,087,500.00	\$ 80,087,500.00	\$ 90,092,400.00	\$100,092,400.00	\$110,092,400.00	\$110,092,400.00	\$110,092,400.00	\$996,897,000.00
Benefits	\$ 50,000,000.00	\$ 36,000,000.00	\$ 85,000.00	\$ 87,500.00	\$ 87,500.00	\$ 87,500.00	\$ 87,500.00	\$ 92,400.00	\$ 92,400.00	\$ 92,400.00	\$ 92,400.00	\$ 92,400.00	\$ 66,897,000.00
Transfer to Project	\$ 20,000,000.00	\$ 30,000,000.00	\$ 40,000,000.00	\$ 50,000,000.00	\$ 60,000,000.00	\$ 70,000,000.00	\$ 80,000,000.00	\$ 90,000,000.00	\$100,000,000.00	\$110,000,000.00	\$110,000,000.00	\$110,000,000.00	\$870,000,000.00
OFFICE COSTS	\$ 11,370.00	\$ 11,770.00	\$ 11,770.00	\$ 11,470.00	\$ 277,480.00	\$ 11,470.00	\$ 11,470.00	\$ 11,470.00	\$ 11,470.00	\$ 11,470.00	\$ 11,770.00	\$ 11,770.00	\$ 138,740.00
Office lease	\$ 9,800.00	\$ 9,800.00	\$ 9,800.00	\$ 9,800.00	\$ 9,800.00	\$ 9,800.00	\$ 9,800.00	\$ 9,800.00	\$ 9,800.00	\$ 9,800.00	\$ 9,800.00	\$ 9,800.00	\$ 117,600.00
Gas	\$ -	\$ 400.00	\$ 400.00	\$ 100.00	\$ 100.00	\$ 100.00	\$ 100.00	\$ 100.00	\$ 100.00	\$ 100.00	\$ 400.00	\$ 400.00	\$ 2,300.00
Electric	\$ 300.00	\$ 300.00	\$ 300.00	\$ 300.00	\$ 300.00	\$ 300.00	\$ 300.00	\$ 300.00	\$ 300.00	\$ 300.00	\$ 300.00	\$ 300.00	\$ 3,600.00
Water	\$ 40.00	\$ 40.00	\$ 40.00	\$ 40.00	\$ 40.00	\$ 40.00	\$ 40.00	\$ 40.00	\$ 40.00	\$ 40.00	\$ 40.00	\$ 40.00	\$ 480.00
Telephone	\$ 250.00	\$ 250.00	\$ 250.00	\$ 250.00	\$ 250.00	\$ 250.00	\$ 250.00	\$ 250.00	\$ 250.00	\$ 250.00	\$ 250.00	\$ 250.00	\$ 3,000.00
Internet access	\$ 180.00	\$ 180.00	\$ 180.00	\$ 180.00	\$ 180.00	\$ 180.00	\$ 180.00	\$ 180.00	\$ 180.00	\$ 180.00	\$ 180.00	\$ 180.00	\$ 2,160.00
Office supplies	\$ 200.00	\$ 200.00	\$ 200.00	\$ 200.00	\$ 200.00	\$ 200.00	\$ 200.00	\$ 200.00	\$ 200.00	\$ 200.00	\$ 200.00	\$ 200.00	\$ 2,400.00
Security	\$ 600.00	\$ 600.00	\$ 600.00	\$ 600.00	\$ 600.00	\$ 600.00	\$ 600.00	\$ 600.00	\$ 600.00	\$ 600.00	\$ 600.00	\$ 600.00	\$ 7,200.00
MARKETING COSTS	\$ 8,100.00	\$ 3,100.00	\$ 3,100.00	\$ 11,100.00	\$ 135,600.00	\$ 3,900.00	\$ 8,100.00	\$ 6,100.00	\$ 3,100.00	\$ 8,100.00	\$ 3,100.00	\$ 6,900.00	\$ 67,800.00
Web site hosting	\$ 500.00	\$ 500.00	\$ 500.00	\$ 500.00	\$ 500.00	\$ 500.00	\$ 500.00	\$ 500.00	\$ 500.00	\$ 500.00	\$ 500.00	\$ 500.00	\$ 6,000.00
Web site updates	\$ 200.00	\$ 200.00	\$ 200.00	\$ 200.00	\$ 200.00	\$ 1,000.00	\$ 200.00	\$ 200.00	\$ 200.00	\$ 200.00	\$ 200.00	\$ 1,000.00	\$ 4,000.00

Figure 15: Restore deleted file revealing Retirement benefits being transferred over a years' time.

6. In a coffee shop, one of Palpatine's employees overheard a conversation he was having with Darth Vader involving a secret meeting place. Is there any evidence supporting the idea that such a meeting place exists?

On the desktop active directory in the suspect image file, we were able to carve hex value pdf file named "BespinTrip", and directions to Bespin International form Jackson Township. (figure 16)

Hex	Text	Application	Source File Metadata	OS Account	Data Artifacts	Analysis Results	Context
Page: 1 of 1		Page		Go to Page: 1		Jump to Offset	
0x00000000:	4C 00 00 00	01 14 02 00	00 00 00 00	C0 00 00 00	L.....		
0x00000010:	00 00 00 46	9B 00 20 00	00 00 00 00	00 00 00 00	...F..		
0x00000020:	00 00 00 00	00 00 00 00	00 00 00 00	00 00 00 00			
0x00000030:	00 00 00 00	00 00 00 00	00 00 00 00	01 00 00 00			
0x00000040:	00 00 00 00	00 00 00 00	00 00 00 00	A8 00 3A 00	?		
0x00000050:	1F 48 3A CC	BF B4 2C DB	4C 42 B0 29	7F E9 9A 87	.H:...,LB.)...		
0x00000060:	C6 41 26 00	01 00 26 00	EF BE 11 00	00 00 5D 57	.A&...&.....]W		
0x00000070:	26 2B FA DD	D9 01 D0 E1	88 6A B8 21	DA 01 36 65	&+.....j.!...6e		
0x00000080:	8C 6A B8 21	DA 01 14 00	6C 00 32 00	00 00 00 00	.j.!...1.2.....		
0x00000090:	00 00 00 00	80 00 42 65	73 70 69 6E	54 72 69 70	BespinTrip		
0x000000a0:	2E 70 64 66	00 00 4E 00	09 00 04 00	EF BE 00 00	.pdf..N.....		
0x000000b0:	00 00 00 00	00 00 2E 00	00 00 00 00	00 00 00 00			
0x000000c0:	00 00 00 00	00 00 00 00	00 00 00 00	00 00 00 00			

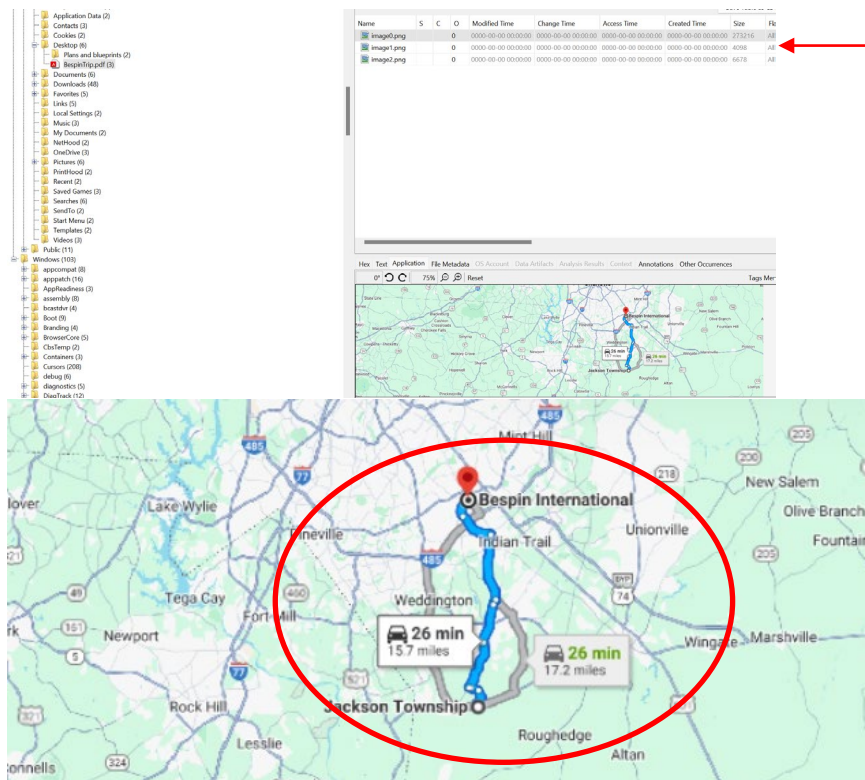


Figure 16: Google Map directions to Bespin carved using Autopsy.

7. Regarding the above evidence, if found when was the file created?

While analyzing the hex data using Autopsy, the first record is the creation time in 26 bytes. Using DCode v5.5, we were able to decode the date the file was created as Friday, 01 December 2023.

The image shows two screenshots. The top screenshot is from the Autopsy hex editor, displaying a hex dump. A red box highlights the hex value `1F 48 3A CC BF B4 2C DB` at offset `0x00000050`. The bottom screenshot is from the DCode v5.5 application. A red arrow points from the highlighted hex value in the top screenshot to the 'Value' field in the 'Value Input' section of the DCode interface. The 'Format' is set to 'Hexadecimal (Little-Endian)'. The 'Value' field contains `1F483ACCBFB42CDB`. Below this, the 'Time Zone' section shows '(UTC-05:00) Eastern Time (US & Canada)'. The 'Date Output' section shows the 'Pattern' as 'dddd, dd MMMM yyyy' and the 'Sample' as 'Friday, 01 December 2023'.

Hex	Text	Application	Source File Metadata	OS Account	Data Artifacts	Analysis Results	Context
Page: 1 of 1	Page	Go to Page: 1	Jump to Offset				
0x00000000:	4C 00 00 00	01 14 02 00	00 00 00 00	C0 00 00 00	L.....		
0x00000010:	00 00 00 46	9B 00 20 00	00 00 00 00	00 00 00 00	...F..		
0x00000020:	00 00 00 00	00 00 00 00	00 00 00 00	00 00 00 00			
0x00000030:	00 00 00 00	00 00 00 00	00 00 00 00	01 00 00 00			
0x00000040:	00 00 00 00	00 00 00 00	00 00 00 00	A8 00 3A 00			
0x00000050:	1F 48 3A CC	BF B4 2C DB	4C 42 B0 29	7F E9 9A 87	.H:...,LB.)...		
0x00000060:	C6 41 26 00	01 00 26 00	EF BE 11 00	00 00 5D 57	.A&...&.....]W		
0x00000070:	26 2B FA DD	D9 01 D0 E1	88 6A B8 21	DA 01 36 65	&+.....j.!..6e		
0x00000080:	8C 6A B8 21	DA 01 14 00	6C 00 32 00	00 00 00 00	.j.!...l.2....		
0x00000090:	00 00 00 00	80 00 42 65	73 70 69 6E	54 72 69 70	BespinTrip		
0x000000a0:	2E 70 64 66	00 00 4E 00	09 00 04 00	EF BE 00 00	.pdf..N.....		
0x000000b0:	00 00 00 00	00 00 2E 00	00 00 00 00	00 00 00 00			
0x000000c0:	00 00 00 00	00 00 00 00	00 00 00 00	00 00 00 00			

DCode v5.5

File Tools Theme Help

Time Decoding Time Encoding

Name	Timestamp
→ Apple Absolute Time (ns) (UTC)	Thursday, 30 November 1916
Apple Absolute Time (ns)	Thursday, 30 November 1916

Value Input

Format: Hexadecimal (Little-Endian)

Value: 1F483ACCBFB42CDB

Decode

Time Zone

Name: (UTC-05:00) Eastern Time (US & Canada)

No Adjustment Select

Date Output

Pattern: dddd, dd MMMM yyyy

Sample: Friday, 01 December 2023

Default

Figure 17: Decoded file creation date using DCode v5.5.

8. Is there any evidence that Palpatine was using digital forensic technology?

By using Autopsy to view the active directory, in the system32 software configuration file, we were able to discover that Palpatine installed FTK Imager v4.7.1.2 and CCleaner v.6.18.(figure 17-18)

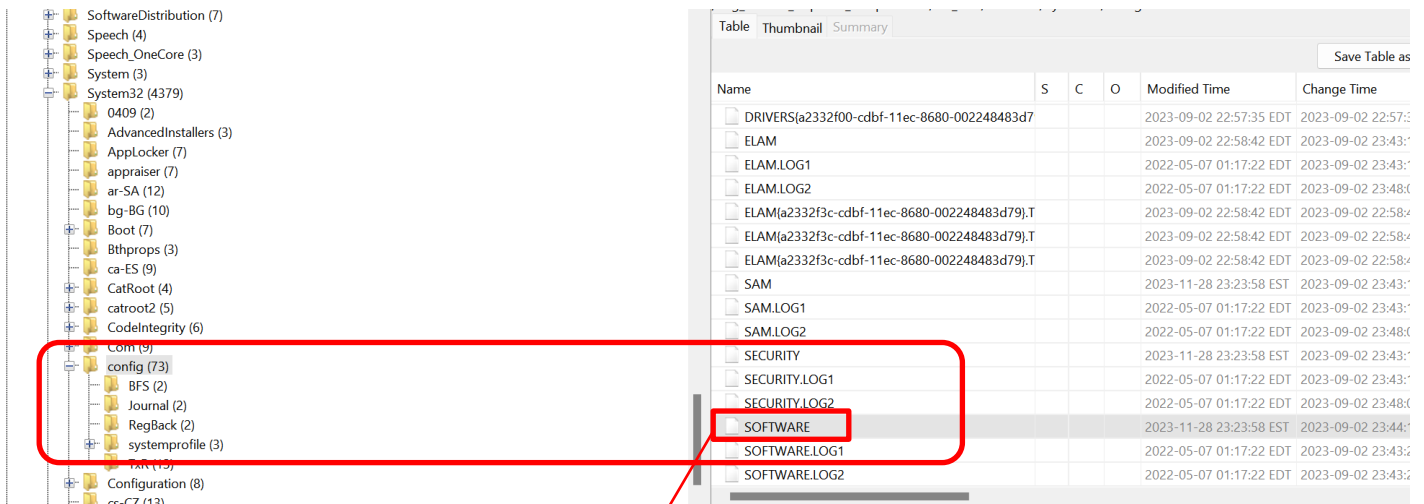


Figure 18: System32 config file captured using Autopsy.

Listing						
Installed Programs						
Table Thumbnail Summary						
Source Name	S	C	O	Program Name	Date/Time	Data Source
SOFTWARE			0	mstsc-4b00-31aa-df6a-4307-9b47-d5cc50009643	2023-11-29 04:23:13 EST	Senator_Palpatine_computer.E01
SOFTWARE			0	CCleaner v.6.18	2023-11-28 05:23:41 EST	Senator_Palpatine_computer.E01
SOFTWARE			0	AccessData FTK Imager v.4.7.1.2	2023-11-28 04:52:20 EST	Senator_Palpatine_computer.E01

Figure 19: Installed software capture using Autopsy.

9. Was Palpatine using any USB devices with this computer?

We were able to discover three USB devices connected to this computer located in the System32 configuration file system. (figure 20)

Hex

Text

Application

Source File Metadata

OS Account

Data Artifacts

Analysis Results

Context

Annotations

Other Occurrences

Result: 2 of 3

Result

←

→

Type	Value
Date/Time	2023-11-28 23:23:59 EST
Device Make	Micron Technology, Inc.
Device Model	Product: 5600
Device ID	2129E329A6F1
Source File Path	/img_Senator_Palpatine_computer.E01/vol_vol7/Windows/System32/config/SYSTEM
Artifact ID	-9223372036854775672

Listing

USB Device Attached

Table

Thumbnail

Summary

Page: 1 of 1

Pages:

←

→

Go to Page:

Source Name	S	C	O	Date/Time	Device Make	Device Model	Device ID	Data Source
 SYSTEM			0	2023-11-28 23:23:59 EST		ROOT_HUB30	4&24054718&0&0	Senator_Palpatine
 SYSTEM			0	2023-11-28 23:23:59 EST	Micron Technology, Inc.	Product: 5600	2129E329A6F1	Senator_Palpatine
 SYSTEM			0	2023-11-28 23:23:59 EST	VirtualBox	USB Tablet	5&12c8f4c0&0&1	Senator_Palpatine

Figure 20: Connected USB devices captured using Autopsy.

10. Palpatine informed investigators that he never touched company resources or money and never viewed information on how to steal from retirement fund. Identify any evidence that may show Palpatine researching how to commit fraud or in the possession of company resources and/or money.

By using Autopsy to view the browser artifacts, we were able to discover that the suspect bookmarked several websites dealing with Payroll fraud, hacking retirement accounts and ways that theft occur at work from employees. (figure 17) Located in the unallocated space of the suspect’s disc image, we were able to recover several images of the subject handing “money”. (figure 18)

Web Bookmarks								
Table Thumbnail Summary								
Source Name	S	C	O	URL	Title	Date Created	Program Name	Do
Bookmarks			1	https://cpahalltalk.com/payroll-fraud/	Payroll Fraud: Easy Ways to Steal - CPA Hall Talk	2023-11-27 23:38:24 EST	Google Chrome	cpa
Bookmarks			1	https://www.investopedia.com/4-major-pension-prob...	3 Ways You Could Lose Your Pension and How to Save...	2023-11-27 23:54:39 EST	Google Chrome	inv
Bookmarks			1	https://thebossmagazine.com/how-do-cybercriminals...	How Do Cybercriminals Hack Retirement Plans?	2023-11-27 23:57:10 EST	Google Chrome	the
Bookmarks			1	https://smallbusiness.chron.com/5-common-ways-e...	5 Most Common Ways Employee Theft Occurs	2023-11-27 23:58:11 EST	Google Chrome	chr
Bookmarks			1	https://www.stevenson.edu/online/about-us/news/di...	Stevenson University	2023-11-27 23:58:57 EST	Google Chrome	ste

Document was last saved: Just now

Figure 21: Web Bookmarks captured using Autopsy.

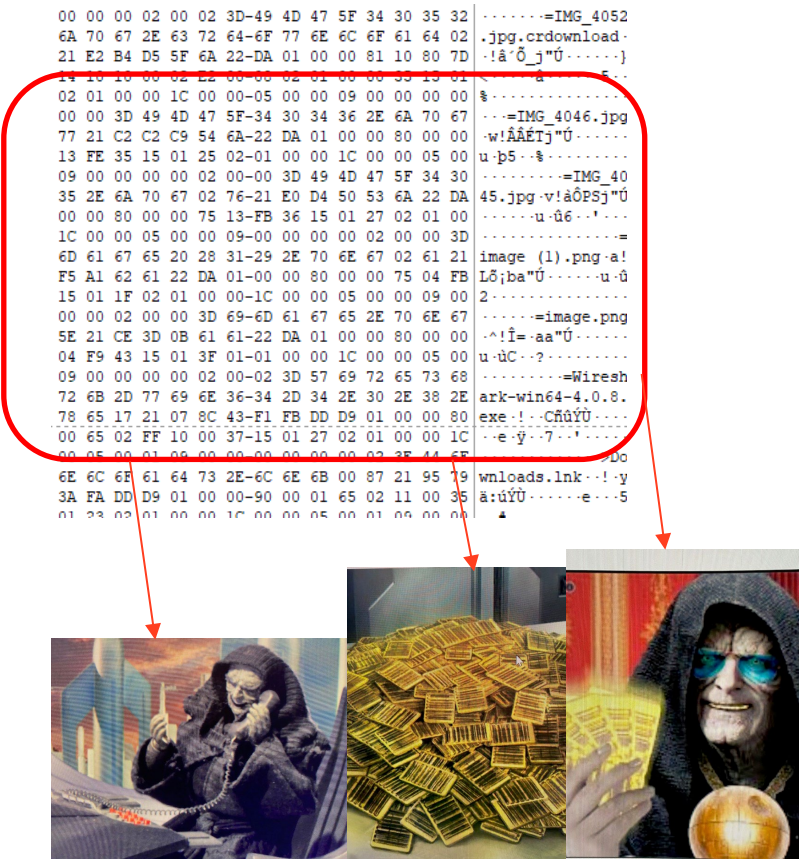


Figure 22: Carved deleted images from hex values using FTK Imager.

11. What are the emails and or usernames of Palpatine and any associates?

From the unallocated space we were able to carve hex values revealing Palpatine's email address as `palpatinedaemperor@gmail.com` and his Emperor Palpatine.

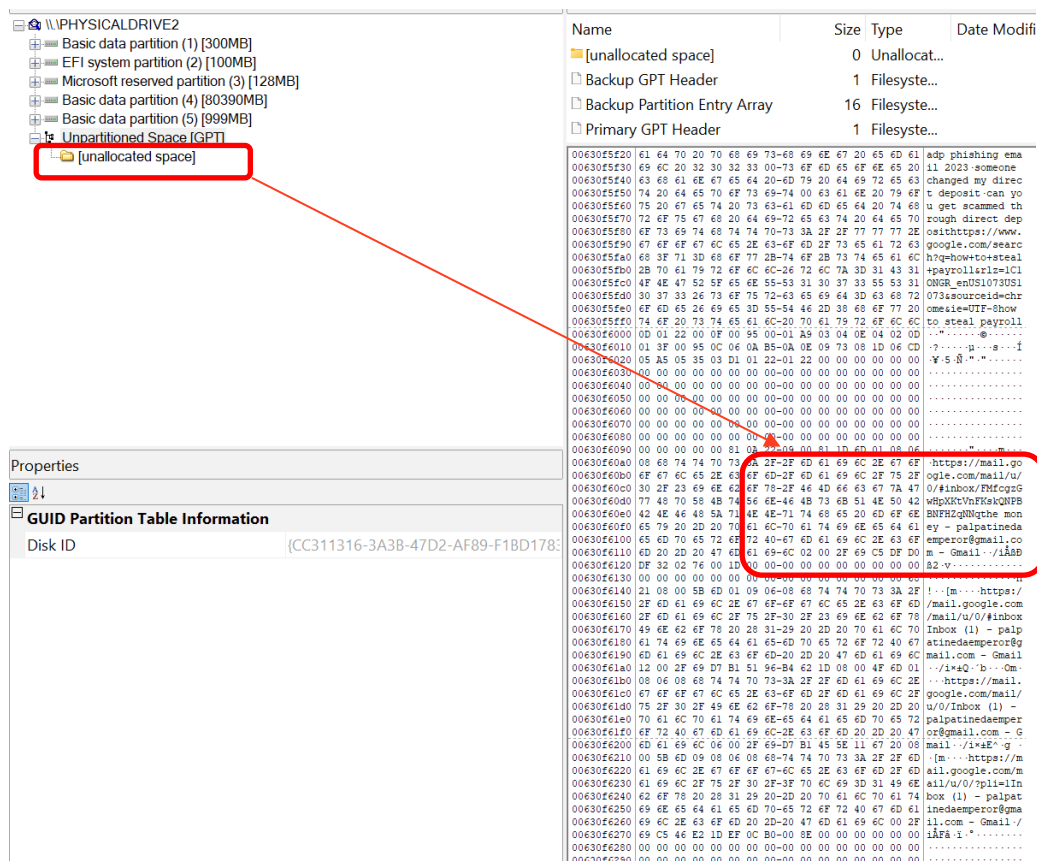


Figure 23: Email address of suspect carved from unallocated space using FTK Imager.

										Save Table as CSV
Source Name	S	C	O	Name	Value	Count	Date Created	Date Accessed	Username	Program Name
 Web Data				43312_361236pi_43312_361236	The empire	1	2023-11-27 23:51:45 EST	2023-11-27 23:51:45 EST	Default	Google Chrome
 Web Data				43312_361227pi_43312_361227	Emperor	1	2023-11-27 23:51:45 EST	2023-11-27 23:51:45 EST	Default	Google Chrome
 Web Data				43312_361230pi_43312_361230	Palpatine	1	2023-11-27 23:51:45 EST	2023-11-27 23:51:45 EST	Default	Google Chrome
 Web Data				43312_361233pi_43312_361233	PalpatineDaEmperor@gmail.com	1	2023-11-27 23:51:45 EST	2023-11-27 23:51:45 EST	Default	Google Chrome
 Web Data				43312_361239pi_43312_361239	Emperor	1	2023-11-27 23:51:45 EST	2023-11-27 23:51:45 EST	Default	Google Chrome
 Web Data				43312_401465pi_43312_401465	Cor	1	2023-11-27 23:51:45 EST	2023-11-27 23:51:45 EST	Default	Google Chrome
 Web Data				43312_401468pi_43312_401468	USA	1	2023-11-27 23:51:45 EST	2023-11-27 23:51:45 EST	Default	Google Chrome
 Web Data				deviceAddress	8284094794	1	2023-11-28 20:01:09 EST	2023-11-28 20:01:09 EST	Default	Google Chrome

Figure 24: Captured email address and username using Autopsy.

12. Was there any evidence found that shows Palpatine may be transferring company funds to another source and images of a “Death Star”?

This evidence artifacts were obtained from the review of the active directory and deleted file fragments. The Excell spreadsheet “Galactic Emp Hire” shows information about Palpatine being the creator of said file. The files contain calculated funds being transfer to a project “Death Star”. (figure 23-26)

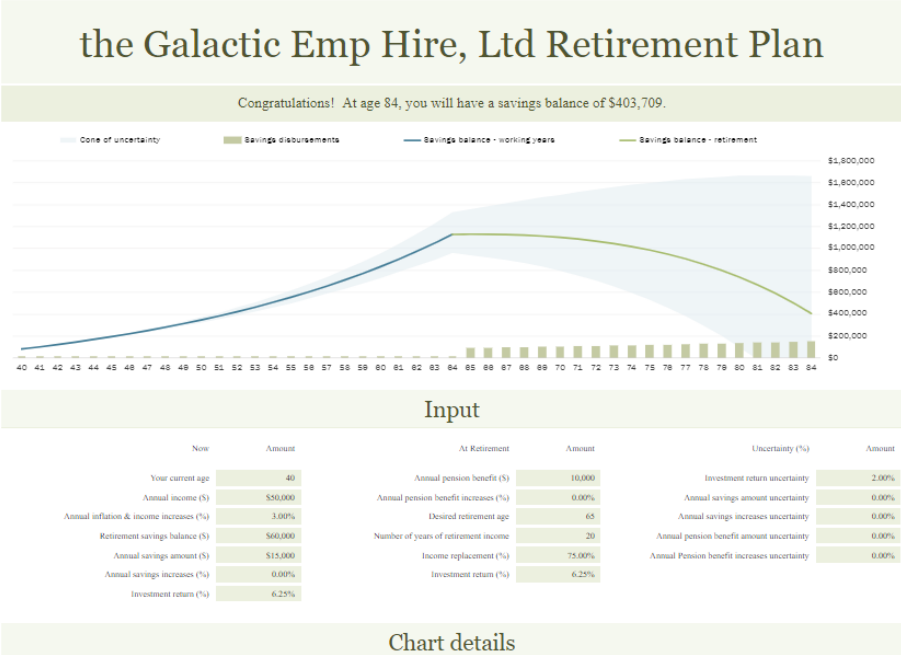


Figure 25: Carved Excel with similar name, file captured from recycle bin using Autopsy.

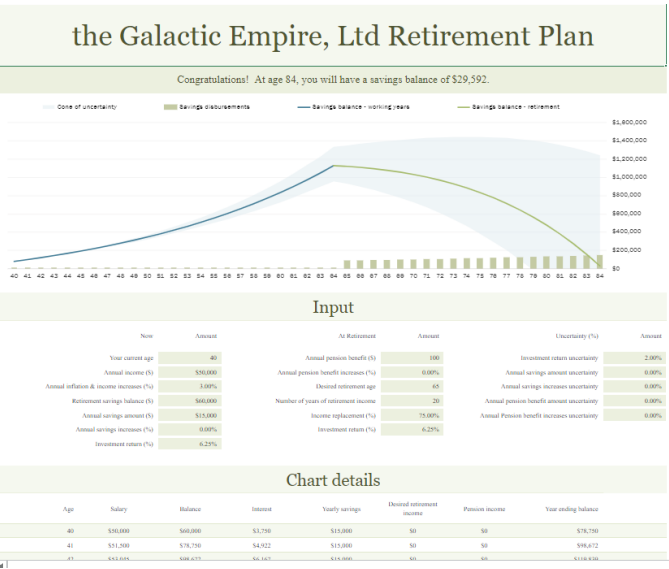


Figure 26: Carved Excel file captured from recycle bin using Autopsy.

Report  Close Case

Listing

/img_Senator_Palpatine_computer.E01/vol_vol7/Users/kostace/Pictures/Screenshots

Table

Thumbnail

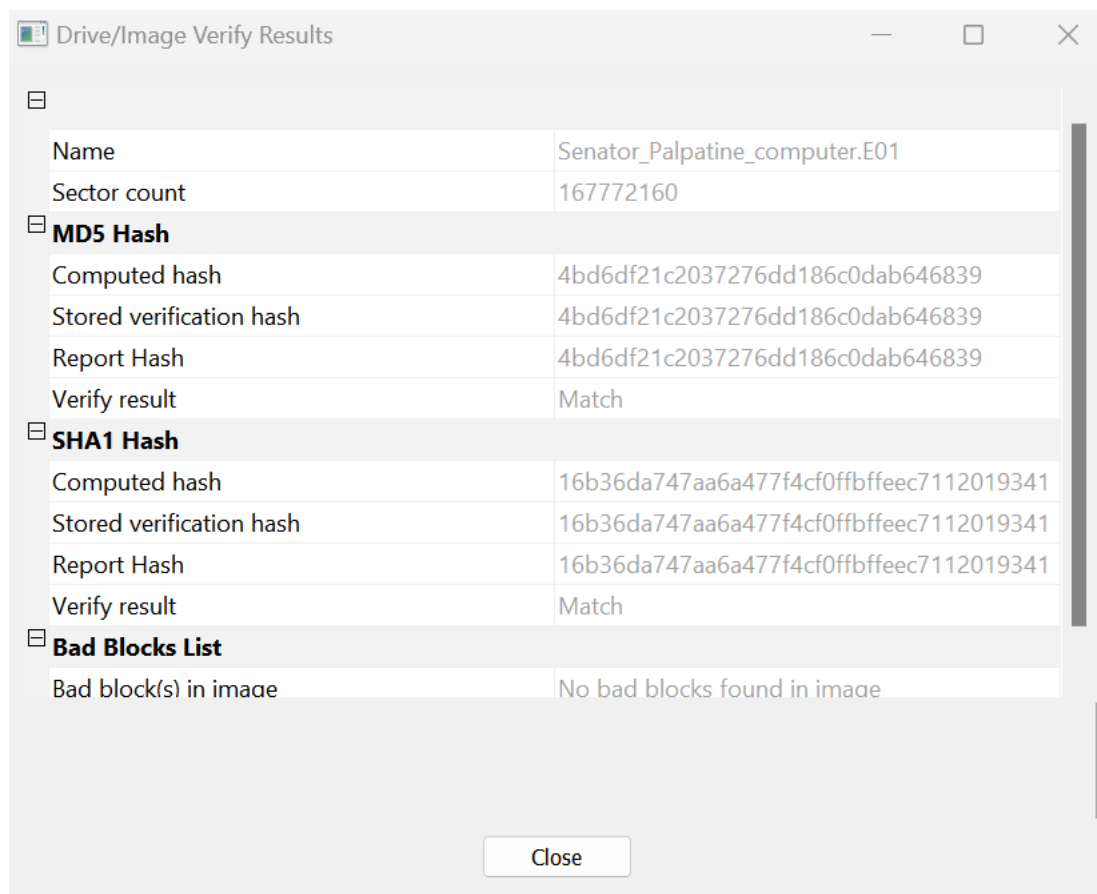
Summary

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Name	S	C	O	Modified Time	Change Time	Access Time	Created Time
 [current folder]				2023-11-28 00:07:58 EST	2023-11-28 00:07:58 EST	2023-11-28 21:30:27 EST	2023-11-28 00:07:58 EST
 [parent folder]				2023-11-28 00:03:13 EST	2023-11-28 00:03:13 EST	2023-11-28 23:09:17 EST	2023-11-28 00:03:13 EST
 desktop.ini			0	2023-11-28 00:03:13 EST	2023-11-28 00:03:13 EST	2023-11-28 23:05:27 EST	2023-11-28 00:03:13 EST
 Lightsaber.png			0	2023-11-28 00:07:58 EST	2023-11-28 00:24:03 EST	2023-11-28 21:23:49 EST	2023-11-28 00:07:58 EST
 Plans for DeathStar.png			0	2023-11-28 00:05:25 EST	2023-11-28 00:24:03 EST	2023-11-28 21:23:53 EST	2023-11-28 00:05:25 EST
 Screenshot 2023-11-28 000313.png			0	2023-11-28 00:03:13 EST	2023-11-28 00:03:13 EST	2023-11-28 21:23:53 EST	2023-11-28 00:03:13 EST
 Screenshot 2023-11-28 000720.png			0	2023-11-28 00:07:20 EST	2023-11-28 00:07:20 EST	2023-11-28 21:23:53 EST	2023-11-28 00:07:20 EST

Figure 27: Images of Death Star found in hidden partition, captured using Autopsy.

Important: The Hash value for each file has been computed to demonstrate that the expert retained the original integrity of the file as imaged from the source computer. No alterations have been made to the individual files.



Verified MD5 Hash Values after analysis.

CONCLUSION

The purpose of this report is to provide corroborative evidence of means, motive and opportunity leading to the fraud perpetrated by Senator Palpatine. The facts presented within this report are those within my area of expertise and knowledge and do not extend to matters outside of the same. To begin my analysis, the images were loaded using Windows 11, 64-bit system with 64 GB RAM, 10.0.22621, I verified the forensic image using FTK Imager 4.7.1.2. All the questions regarding this case were successfully answered. Following this procedure all requested items were satisfied with the utilization of FTK Imager 4.7.1.2, HxD, Autopsy 4.21.0, Arsenal Image Mounter 3.10.262, Volatility3, DCode v5.5. Microsoft Edge Browser, Notepad, and DB Browser for SQLite 3.12.3. The forensic image and answered questions were returned to Detective Kenobi.

Signed:

Kostace Kyser

DFE Kostace Kyser
Digital Forensic Examiner
UNCC Forensic Laboratory